Eden Colors - QR Authenticity System

Project Summary for Project Manager

Document Version: 1.0 **Date:** January 24, 2026 **Project Status:** MVP Complete

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1. Executive Summary

ระบบ QR Authenticity & Activation สำหรับ Eden Colors เป็นระบบติดตามและยืนยันความแท้ของสินค้าความงาม (Filler, Botox, Skincare) ด้วยเทคโนโลยี QR Code ที่มีการเข้ารหัส JWE เพื่อป้องกันการปลอมแปลง

Key Highlights

- **100% Features Complete** - ทุก Sprint ตาม PRD เสร็จสมบูรณ์
- **Full Testing Passed** - ผ่านการทดสอบทุก Flow
- **Bilingual Support** - รองรับ ภาษาไทย และ English
- **Responsive Design** - ใช้งานได้ทุกอุปกรณ์

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2. Feature Overview

Core Features Delivered

Feature	Status	Sprint
User Authentication (JWT)	✅ Complete	Sprint 1
Role-Based Access Control (Admin/Manager/Warehouse)	✅ Complete	Sprint 1
QR Code Generation (JWE Encrypted)	✅ Complete	Sprint 2
Serial Number Auto-Generation (12 digits)	✅ Complete	Sprint 2
Product Verification (Public)	✅ Complete	Sprint 2
GRN (Goods Receipt Note)	✅ Complete	Sprint 3
Outbound with Approval Workflow	✅ Complete	Sprint 3
Manager Approval Board	✅ Complete	Sprint 3
Product Activation (One-Time Lock)	✅ Complete	Sprint 4
PDPA Consent Recording	✅ Complete	Sprint 4
PDF Label Generation (4x6 inch)	✅ Complete	Sprint 4
QR Reprint with Version Control	✅ Complete	Sprint 5
Product Return Management	✅ Complete	Sprint 5
Event Logging & Audit Trail	✅ Complete	Sprint 5
Clinic Management	✅ Complete	Sprint 1
User Management	✅ Complete	Sprint 1
Master Data Management	✅ Complete	Sprint 1

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3. Technical Stack

Frontend

- **Framework:** Next.js 16 (App Router)
- **Language:** TypeScript
- **Styling:** Tailwind CSS v4
- **UI Theme:** Medical Aesthetic (Gold, Charcoal, Mint)

Backend

- **Runtime:** Node.js 18+
- **API:** Next.js Route Handlers
- **Authentication:** JWT with HttpOnly Cookies

Database

- **Primary:** PostgreSQL 15
- **Cache:** Redis (for rate limiting)
- **ORM:** Prisma 6

Security

- **QR Encryption:** JWE (JSON Web Encryption) via JOSE library
- **Password:** bcrypt hashing
- **Session:** Secure HttpOnly cookies
- **Rate Limiting:** Per-IP request limiting

Infrastructure

- **Containerization:** Docker Compose
- **PDF Generation:** jsPDF + qrcode

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4. System Architecture

```
┌──────────────────────────────────────────────────────────┐
│                        FRONTEND                          │
│  Next.js App Router + Tailwind CSS + TypeScript          │
└──────────────────────────────────────────────────────────┘
                              │
                              ▼
┌──────────────────────────────────────────────────────────┐
│                       API LAYER                          │
│  /api/auth      - Authentication                         │
│  /api/public    - Verify & Activate (Public)             │
│  /api/warehouse - GRN, Outbound, Reprint, Return         │
│  /api/manager   - Approval Board                         │
│  /api/admin     - Users, Clinics, Masters, Logs          │
└──────────────────────────────────────────────────────────┘
                              │
                              ▼
┌──────────────────────────────────────────────────────────┐
│                      DATA LAYER                          │
│  PostgreSQL (Primary) + Redis (Cache)                    │
│  Prisma ORM                                              │
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```

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5. User Roles & Permissions

Role	Permissions
Admin	Full system access, User/Clinic management, Master data, Event logs
Manager	Approve/Reject outbound requests, View reports, Limited product access
Warehouse	GRN creation, Outbound creation, Print labels, Reprint QR, Product returns
Public	Verify QR (no login), Activate products (no login)

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6. Business Workflow

```
┌──────────────┐    ┌──────────────┐    ┌──────────────┐    ┌──────────────┐
│   Inbound    │ → │  Outbound    │ → │   Approve    │ → │   Verify     │
│    (GRN)     │    │   Create     │    │  (Manager)   │    │  (Public)    │
└──────────────┘    └──────────────┘    └──────────────┘    └──────────────┘
                                                                   │
                                                                   ▼
                                                             ┌──────────────┐
                                                             │  Activate    │
                                                             │  (Customer)  │
                                                             └──────────────┘


Product Status Flow:
IN_STOCK → PENDING_OUT → SHIPPED → ACTIVATED
                ↓                      ↓
            (Reject)               RETURNED
                ↓
            IN_STOCK
```

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7. Test Results Summary

Automated Testing

Test Category	Tests	Pass	Fail
Authentication	7	7	0
GRN Flow	4	4	0
Outbound Flow	3	3	0
Activation Lock	3	3	0
Reprint Versioning	5	5	0
Return Flow	4	4	0
PDF Generation	3	3	0
Total	**29**	**29**	**0**

Pass Rate: 100%

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8. Test Accounts

Role	Username	Password
Admin	admin	admin123
Warehouse	warehouse1	warehouse123
Manager	manager1	manager123

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9. Known Issues & Limitations

Current Limitations

1. **Thai Font in PDF** - ฟอนต์ไทยใน PDF อาจแสดงผลไม่ถูกต้องในบางกรณี (fallback to English)
2. **Rate Limiting** - In-memory rate limiting (resets on server restart)
3. **Image Storage** - ไม่มี image upload ในระบบปัจจุบัน

Future Enhancements (Out of MVP Scope)

1. Dashboard Analytics & Reports
2. Batch QR Printing Optimization
3. Push Notifications for Approvals
4. Mobile App (React Native)
5. Integration with ERP/Accounting Systems
6. Advanced Search & Filtering

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10. Deployment Checklist

Pre-Production

- [ ] Change JWT_SECRET to production key
- [ ] Configure production PostgreSQL
- [ ] Configure production Redis
- [ ] Set up SSL certificates
- [ ] Configure domain DNS
- [ ] Set up backup strategy

Environment Variables Required

```
DATABASE_URL=postgresql://...
REDIS_URL=redis://...
JWT_SECRET=<strong-secret-key>
NEXT_PUBLIC_BASE_URL=https://your-domain.com
```

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11. Project Timeline

Phase	Duration	Status
Sprint 0: Setup	1 day	✅ Complete
Sprint 1: Auth/Admin	1 day	✅ Complete
Sprint 2: QR/Verify	1 day	✅ Complete

Sprint 3: GRN/Outbound	2 days	✅ Complete
Sprint 4: Activation/PDF	1 day	✅ Complete
Sprint 5: Reprint/Return	1 day	✅ Complete
Testing & UAT	1 day	✅ Complete
UI Design Polish	1 day	✅ Complete

Total Development Time: ~9 days

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12. Support & Maintenance

Documentation Available

- `docs/USER_GUIDE.md` - Full system user guide
- `docs/CUSTOMER_GUIDE.md` - End-user guide for customers
- `CLAUDE.md` - Developer quick reference
- `README.md` - Project setup instructions

Key Commands

```
npm run dev      # Start development server
npm run build    # Build for production
npm run db:up    # Start database containers
npm run db:seed  # Seed test data
```

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13. Conclusion

โปรเจค Eden Colors QR Authenticity System ได้พัฒนาเสร็จสมบูรณ์ตาม PRD requirements ทั้งหมด พร้อมสำหรับ:

1. **User Acceptance Testing (UAT)** - ทดสอบกับผู้ใช้จริง
2. **Production Deployment** - Deploy ไปยัง production server
3. **Training** - อบรมการใช้งานให้กับทีม

ระบบมีความพร้อมในการป้องกันสินค้าปลอม ติดตามสินค้าตลอด supply chain และสร้างความเชื่อมั่นให้กับลูกค้า

Prepared by: Development Team **Approved by:** _____ **Date:** _____